

Practical Exercise

Google Big Query Dataset: Retail Sales Dataset

QUESTIONS:

Q1 WHERE Clause

The screenshot shows the Google BigQuery console interface. The left sidebar contains navigation options: Overview, Studio (selected), Agents, Pipelines and integration, Data transfers, Dataform, Scheduled queries, Scheduling, Governance, Sharing (Analytics Hub), Policy tags, Metadata curation, Administration, Monitoring, Partner Centre, Settings, and Release notes. The main area displays 'Question 1' with a SQL query:

```
1 SELECT *
2 FROM `sql-big-query-494420.retail_data.retail_sales`
3 WHERE EXTRACT(YEAR FROM Date) = 2023;
4
5
```

Below the query, a status message indicates: 'This query will process 72.69 KB when run. Using on-demand processing quota'. The 'Query results' section shows a table with 10 rows and 9 columns: Row, Transaction ID, Date, Customer ID, Gender, Age, Product Category, and Quantity. The results are as follows:

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity
1	191	2023-10-18	CUST1191	Male	64	Beauty	
2	204	2023-09-28	CUST204	Male	39	Beauty	
3	230	2023-04-23	CUST230	Male	54	Beauty	
4	232	2023-02-06	CUST232	Female	43	Beauty	
5	309	2023-12-23	CUST309	Female	26	Beauty	
6	310	2023-10-12	CUST310	Female	28	Beauty	
7	363	2023-06-03	CUST363	Male	64	Beauty	
8	371	2023-02-21	CUST371	Female	20	Beauty	
9	397	2023-03-10	CUST397	Female	30	Beauty	
10	454	2023-02-22	CUST454	Female	46	Beauty	

At the bottom, the 'Job history' section is visible with a 'Show' button. The 'Results per page' is set to 50, showing 1 - 50 of 998 results.

Q2 Filtering + Conditions

The screenshot shows the Google Cloud BigQuery console interface. The top navigation bar includes the Google Cloud logo and a search bar. The left sidebar contains the Explorer panel with options like Home, Starred, Shared with me, Job history, and a list of datasets, including 'sql-big-query-494420'. The main area displays 'Question 2' with a SQL query and its results.

SQL Query:

```
1 SELECT *
2 FROM `sql-big-query-494420.retail_data.retail_sales`
3 WHERE `Total Amount` > (
4   SELECT AVG(`Total Amount`)
5   FROM `sql-big-query-494420.retail_data.retail_sales`
6 );
```

Query results:

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quant
1	21	2023-01-14	CUST021	Female	50	Beauty	
2	28	2023-04-23	CUST028	Female	43	Beauty	
3	128	2023-07-05	CUST128	Male	25	Beauty	
4	220	2023-03-03	CUST220	Male	64	Beauty	
5	238	2023-01-17	CUST238	Female	39	Beauty	
6	364	2023-08-23	CUST364	Female	19	Beauty	
7	408	2023-04-15	CUST408	Female	64	Beauty	
8	537	2023-06-03	CUST537	Female	21	Beauty	
9	539	2023-06-08	CUST539	Male	25	Beauty	
10	541	2023-07-20	CUST541	Male	66	Beauty	

Results per page: 50 | 1 - 50 of 350 | < > >|

Q3 Aggregate Functions

Anchor of Hope x big query - Google Search x Datasets - BigQuery - SQL B x brightlearn portal - Google S x BrightLearn x +

console.cloud.google.com/bigquery?facet_url=https%3F%2Fcloud.google.com%2Ffree&project=sql-big-query-494420&ws=1m281m612m51m31ssql-big-query-494420...

Gmail YouTube Adobe Acrobat Marketing, Sales &... Maps All Bookmarks

Google Cloud SQL Big Query Search (/) for resources, docs, products and more Search

Question 1 x Question 2 x Question 3 x +

Explorer + Add data

Search for resources

Home
Starred
Shared with me
Job history

sql-big-query-494420

Datasets
Connections
Conversations
Agents
Queries
(Classic) Queries
Notebooks
Data canvases
Data preparations
Pipelines
Repositories

1 SELECT SUM('Total Amount') AS Total_Revenue
2 FROM 'sql-big-query-494420.retail_data.retail_sales';
3

Run Save query Download Share Schedule Open in More

This query will process 7.81 KB when run.
Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualisation JSON Execution details Execution graph

Row	Total_Revenue
1	456000

Results per page: 50 1 - 1 of 1 < >

Job history Show

Q4 DISTINCT

Anchor of Hope x big query - Google Search x Datasets - BigQuery - SQL B x brightlearn portal - Google S x BrightLearn x +

console.cloud.google.com/bigquery?facet_url=https%3F%2Fcloud.google.com%2Ffree&project=sql-big-query-494420&ws=1m281m612m51m31ssql-big-query-494420...

Gmail YouTube Adobe Acrobat Marketing, Sales &... Maps All Bookmarks

Google Cloud SQL Big Query Search (/) for resources, docs, products and more Search

Question 1 x Question 2 x Question 3 x Question 4 x +

Explorer + Add data

Search for resources

Home
Starred
Shared with me
Job history

sql-big-query-494420

Datasets
Connections
Conversations
Agents
Queries
(Classic) Queries
Notebooks
Data canvases
Data preparations
Pipelines
Repositories

1 SELECT DISTINCT 'Product Category'
2 FROM 'sql-big-query-494420.retail_data.retail_sales';
3

Run Save query Download Share Schedule Open in More

This query will process 10.17 KB when run.
Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualisation JSON Execution details Execution graph

Row	Product Category
1	Beauty
2	Clothing
3	Electronics

Results per page: 50 1 - 3 of 3 < >

Job history Show

Q5 GROUP BY

Anchor of Hope x big query - Google Search x Datasets - BigQuery - SQL B x brightlearn portal - Google S x BrightLearn x +

console.cloud.google.com/bigquery?facet_url=https:%2F%2Fcloud.google.com%2Ffree&project=sql-big-query-494420&ws=1m351m6112m51m311ssql-big-query-494420...

Gmail YouTube Adobe Acrobat Marketing, Sales & Maps All Bookmarks

Google Cloud SQL Big Query Search (/) for resources, docs, products and more Search

Question 1 x Question 2 x Question 3 x Question 4 x Question 5 x + -

Explorer + Add data

Search for resources

Home

Starred

Shared with me

Job history

sql-big-query-494420

Datasets

Connections

Conversations

Agents

Queries

(Classic) Queries

Notebooks

Data canvases

Data preparations

Pipelines

Repositories

Question 5 Run Save query Download Share Schedule Open in More

```
1 SELECT 'Product Category',
2       SUM(Quantity) AS Total_Quantity
3 FROM `sql-big-query-494420.retail_data.retail_sales`
4 GROUP BY 'Product Category';
5
```

This query will process 17.98 KB when run.
Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualisation JSON Execution details Execution graph

Row	Product Category	Total_Quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

Results per page: 50 1 - 3 of 3 |< < > >|

Job history Show

Q6 CASE Statement

Anchor of Hope x big query - Google Search x Datasets - BigQuery - SQL B x brightlearn portal - Google S x BrightLearn x +

console.cloud.google.com/bigquery?facet_url=https:%2F%2Fcloud.google.com%2Ffree&project=sql-big-query-494420&ws=1m491m6112m51m311ssql-big-query-494420...

Gmail YouTube Adobe Acrobat Marketing, Sales & Maps All Bookmarks

Google Cloud SQL Big Query Search (/) for resources, docs, products and more Search

Question 2 x Question 3 x Question 4 x Question 5 x Question 6 x Question 7 x + -

Explorer + Add data

Search for resources

Home

Starred

Shared with me

Job history

sql-big-query-494420

Datasets

Connections

Conversations

Agents

Queries

(Classic) Queries

Notebooks

Data canvases

Data preparations

Pipelines

Repositories

Question 6 Run Save query Download Share Schedule Open in More

```
1 SELECT 'Customer ID',
2       Age,
3       CASE
4         WHEN Age < 30 THEN 'Youth'
5         WHEN Age BETWEEN 30 AND 59 THEN 'Adult'
6         ELSE 'Senior'
7       END AS Age_Group
8 FROM `sql-big-query-494420.retail_data.retail_sales`;
9
```

This query will process 16.6 KB when run.

Query results Chat Save results Open in

Job information Results Visualisation JSON Execution details Execution graph

Row	Customer ID	Age	Age_Group
1	CUST191	64	Senior
2	CUST204	39	Adult
3	CUST230	54	Adult
4	CUST232	43	Adult
5	CUST309	26	Youth
6	CUST310	28	Youth
7	CUST363	64	Senior
8	CUST371	20	Youth

Results per page: 50 1 - 50 of 1000 |< < > >|

Job history Show

Q7 Conditional Aggregation

Anchor of Hope x big query - Google Search x Datasets - BigQuery - SQL B x brightlearn portal - Google S x BrightLearn x +

console.cloud.google.com/bigquery?facet_url=https:%2F%2Fcloud.google.com%2Ffree&project=sql-big-query-494420&ws=1m49i1m6i12m5i1m3i1ssql-big-query-494420...

Gmail YouTube Adobe Acrobat Marketing, Sales &... Maps All Bookmarks

Google Cloud SQL Big Query Search (/) for resources, docs, products and more Search

Question 2 x Question 3 x Question 4 x Question 5 x Question 6 x Question 7 x +

Explorer + Add data Search for resources

Home Starred Shared with me Job history

sql-big-query-494420

Datasets Connections Conversations Agents Queries (Classic) Queries Notebooks Data canvases Data preparations Pipelines Repositories

Question 7 Run Save query Download Share Schedule Open in More

```
1 SELECT Gender,
2     COUNTIF('Total Amount' > 500) AS High_Value_Transactions
3 FROM `sql-big-query-494420.retail_data.retail_sales`
4 GROUP BY Gender;
```

This query will process 14.67 KB when run.
Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualisation JSON Execution details Execution graph

Row	Gender	High_Value_Trans...
1	Male	144
2	Female	155

Results per page: 50 1 - 2 of 2 |< < > >|

Job history Show

Q8 HAVING Clause

Anchor of Hope x big query - Google Search x Datasets - BigQuery - SQL B x brightlearn portal - Google S x BrightLearn x +

console.cloud.google.com/bigquery?facet_url=https:%2F%2Fcloud.google.com%2Ffree&project=sql-big-query-494420&ws=1m49i1m6i12m5i1m3i1ssql-big-query-494420...

Gmail YouTube Adobe Acrobat Marketing, Sales &... Maps All Bookmarks

Google Cloud SQL Big Query Search (/) for resources, docs, products and more Search

Question 3 x Question 4 x Question 5 x Question 6 x Question 7 x Question 8 x +

Explorer + Add data Search for resources

Home Starred Shared with me Job history

sql-big-query-494420

Datasets Connections Conversations Agents Queries (Classic) Queries Notebooks Data canvases Data preparations Pipelines Repositories

Question 8 Run Save query Download Share Schedule Open in More

```
1 SELECT 'Product Category',
2     SUM('Total Amount') AS Total_Revenue
3 FROM `sql-big-query-494420.retail_data.retail_sales`
4 GROUP BY 'Product Category'
5 HAVING SUM('Total Amount') > 5000;
```

This query will process 17.98 KB when run.
Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualisation JSON Execution details Execution graph

Row	Product Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Results per page: 50 1 - 3 of 3 |< < > >|

Job history Show

Q9 Calculated Fields

Anchor of Hope x big query - Google Search x Datasets - BigQuery - SQL B x brightlearn portal - Google S x BrightLearn x +

console.cloud.google.com/bigquery?facet_url=https:%2F%2Fcloud.google.com%2Ffree&project=sql-big-query-494420&ws=1m631m6112m51m311ssql-big-query-494420...

Gmail YouTube Adobe Acrobat Marketing, Sales &... Maps All Bookmarks

Google Cloud SQL Big Query Search (/) for resources, docs, products and more Search

Question 4 x Question 5 x Question 6 x Question 7 x Question 8 x Question 9 x

Explorer + Add data Search for resources

Home Starred Shared with me Job history

sql-big-query-494420

Datasets Connections Conversations Agents Queries (Classic) Queries Notebooks Data canvases Data preparations Pipelines Repositories

Question 9 Run Save query Download Share Schedule Open in More

```

1 SELECT 'Transaction ID',
2       'Price per Unit',
3       CASE
4         WHEN 'Price per Unit' < 50 THEN 'Cheap'
5         WHEN 'Price per Unit' BETWEEN 50 AND 200 THEN 'Moderate'
6         ELSE 'Expensive'
7       END AS Unit_Cost_Category
8 FROM `sql-big-query-494420.retail_data.retail_sales`;
9

```

This query will process 15.63 KB when run.
Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualisation JSON Execution details Execution graph

Row	Transaction ID	Price per Unit	Unit_Cost_Category
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap
4	232	25	Cheap
5	309	25	Cheap
6	310	25	Cheap
7	243	25	Cheap

Results per page: 50 1 - 50 of 1000

Job history Show

Q10 Combining WHERE + CASE

Anchor of Hope x big query - Google Search x Datasets - BigQuery - SQL B x brightlearn portal - Google S x BrightLearn x +

console.cloud.google.com/bigquery?facet_url=https:%2F%2Fcloud.google.com%2Ffree&project=sql-big-query-494420&ws=1m701m6112m51m311ssql-big-query-494420...

Gmail YouTube Adobe Acrobat Marketing, Sales &... Maps All Bookmarks

Google Cloud SQL Big Query Search (/) for resources, docs, products and more Search

Question 5 x Question 6 x Question 7 x Question 8 x Question 9 x Question 10 x

Explorer + Add data Search for resources

Home Starred Shared with me Job history

sql-big-query-494420

Datasets Connections Conversations Agents Queries (Classic) Queries Notebooks Data canvases Data preparations Pipelines Repositories

Question 10 Run Save query Download Share Schedule Open in More

```

1 SELECT 'Customer ID',
2       'Age',
3       'Total Amount',
4       CASE
5         WHEN 'Total Amount' > 1000 THEN 'High'
6         ELSE 'Low'
7       END AS Spending_Level
8 FROM `sql-big-query-494420.retail_data.retail_sales`
9 WHERE Age >= 40;
10

```

This query will process 24.42 KB when run.
Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualisation JSON Execution details Execution graph

Row	Customer ID	Age	Total Amount	Spending_Level
1	CUST191	64	25	Low
2	CUST230	54	25	Low
3	CUST232	43	25	Low
4	CUST363	64	25	Low
5	CUST454	46	25	Low
6	CUST512	57	25	Low
7	CUST791	51	25	Low

Results per page: 50 1 - 50 of 558

Job history Show